Question

The transition metal center $\mathbf{A}$ located in the d-block or ds-block can form a complex structure of the form $[\mathbf{AB}_6\mathbf{C}_{12}]$, where the order of its point group is 24, and there are no C_4 or S_4 axes present. Here, $\mathbf{A, B, C}$ are atoms and can be treated as points. All bond angles in this complex are non-acute, and the distances between symmetry-independent $\mathbf{C-C}$ atom pairs are pairwise distinct. Determine the number of faces of the convex polyhedron formed by the outermost atoms as vertices in this complex molecule.

A. 2

B. 3

C. 4

D. 6

E. 8

F. 9

G. 10

H. 12

I. 14

J. 15

K. 18

L. 20

M. 24

N. 30

O. 45

P. 48

Q. 55

R. 60

S. 66

T. 72

Answer

Correct Answer: L

Detailed Explanation

The point groups of order 24 include

$$C_{24}, C_{12\mathrm{v}}, C_{12\mathrm{h}}, D_{12}, D_{6\mathrm{h}}, D_{6\mathrm{d}}, S_{24}, T_{\mathrm{h}}, T_{\mathrm{d}}, O$$

and others. Among these, only the two point groups $D_{6\mathrm{h}}$ and T_{h} satisfy the condition of having no C_4 or S_4 axis.

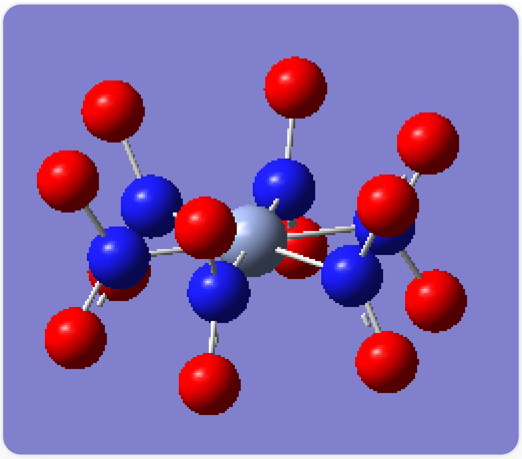
CHECKPOINT 1 PTS

Only the point groups $D_{6\mathrm{h}}$ and T_{h} satisfy the condition

If the point group is $D_{6\mathrm{h}}$, then with $\mathbf{A}$ at the center, six $\mathbf{B}$ atoms form a regular planar hexagon, with each $\mathbf{B}$ atom connected to two $\mathbf{C}$ atoms. The $\mathbf{C}$ atoms are symmetrically distributed above and below the plane, forming a convex polyhedron that is a hexagonal prism.

CHECKPOINT 1 PTS

If the point group is D_{6h}, its structure is a hexagonal prism



The molecular structure of D_{6h}, featuring 1 silver sphere at the center, 6 blue spheres connected to the silver sphere forming a planar hexagon, with each blue sphere bonded to two red spheres symmetrically distributed on both sides of the plane

However, since d-block and ds-block elements lack available f orbitals for bonding, they cannot form a regular planar hexagon coordination. Therefore, the point group of this complex is not $D_{6\mathrm{h}}$.

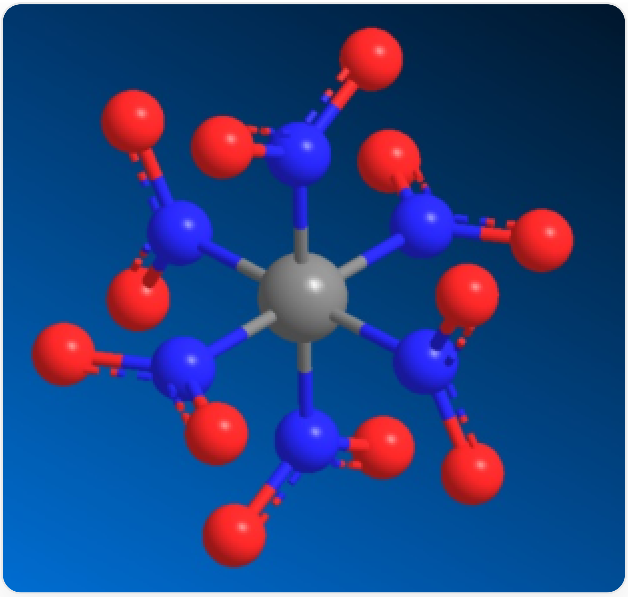
CHECKPOINT 1 PTS

d-block and ds-block elements lack available f orbitals for bonding

On the other hand, if the point group is T_{h}, then with $\mathbf{A}$ at the center, six $\mathbf{B}$ atoms form a regular octahedron, with each $\mathbf{B}$ atom connected to two $\mathbf{C}$ atoms. The $\mathbf{BC}_2$ units lie within the $\mathbf{AB}_4$ cross-section of the octahedron, and the planes of adjacent $\mathbf{BC}_2$ units are mutually perpendicular. The outermost 12 $\mathbf{C}$ atoms form the shape of a triangular icosahedron.

CHECKPOINT 2 PTS

If the point group is T_h, its structure is a triangular icosahedron



The molecular structure of T_h, featuring 1 silver sphere at the center, 6 blue spheres forming a regular octahedron with red spheres, each blue sphere bonded to 2 red spheres. The red-blue-red units lie within the plane formed by 1 silver sphere and 4 blue spheres, with adjacent blue spheres having their red-blue-red planes mutually perpendicular

The triangular icosahedron has 20 triangular faces.

CHECKPOINT 1 PTS

There are 20 triangular faces